

ANALYTICAL BASED QUESTION

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1. Two algorithm solve the same problem. Algorithm A uses recursion, while Algorithm B uses an explicit stack. compare both approach in term of memory usage, execution flow and risk of stack overflow.

A. COMPARE RECURSION AND EXPLICIT STACK

Two algorithm A uses problems

- Algorithm A uses Recursion
- Algorithm B uses an Explicit Stack

compare both approach in terms memory usage, execution, and risk of stack overflow.

<u>feature</u>	<u>Recursion</u>	<u>Explicit Stack</u>
memory usage	uses the system call stack automatically	uses a stack created by the programmer in memory.
Execution flow	function calls itself Repeatedly until case is Reached.	uses push and pop operation inside loops to control manually
Control	less control because the compiler manages stack	more control because the programmer manages.
performance	Slightly slower due to function call overhead	usually faster because it avoids Recursion function

Advantages of Recursion:

- Simple and easy to understand.
- Suitable for tree traversal divide-and-conquer algorithm and graph algorithm.

Advantage of explicit stack:

- Better control over memory
- Suitable for large dataset
- Avoid Recursion limit and stack overflow.

Conclusion:

Recursion is easier to write and understand but an explicit stack provide memory management and Reduces the risk stack overflow.

2. A stack has a capacity of 5 element. Explain what happen if following operation executed.

push(A), push(A), push(C), push(D), push(E), push(F)

Identify the error and suggest two possible solution.

- A. A stack has a capacity of 5 element.

OPERATION PERFORMED:

PUSH (A) PUSH (D)

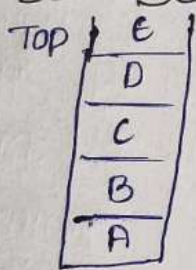
PUSH (B) PUSH (E)

PUSH (C) PUSH (F)

STEP BY STEP EXECUTION

<u>operation</u>	<u>STACK CONTENT</u>	<u>STATUS</u>
push (A)	A	Success
push (B)	A, B	Success
push (C)	A, B, C	Success
push (D)	A, B, C, D	Success
push (E)	A, B, C, D, E	Success (Stack Full)
push (F)	A, B, C, D, E, F	Error: Stack Overflow.

STACK DIAGRAM:



capacity = 5

STACK OVERFLOW:

A stack overflow occurs when a push operation is performed on stack that already reaches its maximum capacity.

Two possible solution:

Solution 1: Increase stack size

- Increase the capacity stack so it can store more element

Solution 2: Check before push

- Before inserting a new element, check whether stack full
- If it is full, display a message like.

conclusion:

Since the stack capacity is 5, only A, B, C, D and E can be
The push (E) operation causes a stack overflow because
there is no space left in the stack.

